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Application progress of machine learning in patient-ventilator asynchrony during mechanical ventilation: a systematic review

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Abstract

Introduction Patient-ventilator asynchrony (PVA) is a common and harmful complication during mechanical ventilation, often requiring labor-intensive manual assessment. Machine learning (ML) offers a promising approach for automated and accurate PVA detection and prediction. We conducted a systematic review to evaluate the methodologies and performance of ML models applied to PVA.

Methods Following a pre-established PROSPERO-registered protocol (CRD42024626163), we systematically searched PubMed, Embase, Web of Science, and Cochrane databases from inception to November 2024. Original studies applying ML techniques to assess PVA in adult patients receiving mechanical ventilation were included. A refined quality assessment method was developed to evaluate methodological and technical features of included studies.

Results Among the 74 articles, fourteen were included, employing 11 distinct ML methods to address 19 targeted types of PVA. Thirteen studies focused on PVA identification, while only one predicted its occurrence. ML models demonstrated promising performance for PVA identification, with high F1 scores (ranging from 0.731 to 0.988) and strong results across other robust metrics (e.g., accuracy, sensitivity, specificity, MCC, AUROC) where reported. However, only two studies conducted external validation. Furthermore, reporting of crucial technical details, such as class imbalance handling (addressed in 5 studies) and hyperparameter tuning (described in 4 studies), was often limited.

Conclusions ML models show significant potential for identifying and predicting PVA, as evidenced by their high reported performance metrics. However, their widespread clinical application is currently limited by substantial methodological heterogeneity, variability in PVA definitions and outcome measures, insufficient external validation, and inconsistent reporting of crucial technical details. Future research should prioritize developing generalizable models validated externally in diverse clinical settings, standardizing PVA definitions and outcome measures, and providing detailed methodological reporting to enhance reproducibility and clinical applicability.

Keywords Mechanical ventilation, Patient-ventilator asynchrony, Machine learning, Artificial intelligence, Systematic review

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Introduction

Mechanical ventilation (MV) serves as a vital life-support intervention in the Intensive Care Unit (ICU), providing stable respiratory assistance to patients by maintaining oxygenation, alleviating respiratory muscle fatigue, and mitigating respiratory distress [1]. It plays a pivotal role in the management of critically ill patients. Treatment typically commences with controlled ventilation, where respiration is entirely regulated by the ventilator. As patients regain their autonomous breathing capacity, the therapeutic approach transitions to assisted ventilation. In this case, the interaction between the patient and the ventilator constitutes a critical element of the MV process, necessitating that the ventilator responds appropriately to the patient's respiratory demands. Any discrepancy between the patient's respiratory needs and the support delivered by the ventilator during MV is termed patient-ventilator asynchrony (PVA).

Among patients undergoing MV while retaining spontaneous breathing, the reported incidence of patient-ventilator asynchrony (PVA) is as high as 10-85% [2]. Frequent occurrences of PVA have been implicated in the development of ventilator-associated lung injury (VILI) [3], failed weaning from mechanical ventilation, prolonged ventilation duration, and increased mortality [4]. Currently, PVA monitoring primarily relies on real-time visual inspection of respiratory waveforms by skilled clinicians [5]. While fundamental, this method is labor-intensive, time-consuming, and impractical for comprehensive, continuous bedside assessment across the tens of thousands of respiratory cycles patients experience daily [6]. More advanced physiological modalities, such as electrical activity of the diaphragm (Edi) and esophageal pressure (Pes) measurement, offer higher accuracy in detecting specific PVA types by directly assessing patient respiratory effort [7]. However, these techniques are invasive, require specialized equipment and expertise, and are not universally available or suitable for continuous, automated monitoring. Furthermore, ventilator-integrated rule-based methods (RBMs), though offering some automation, are limited by fixed thresholds and lack adaptability to diverse patient conditions or novel asynchrony patterns [8, 9].

For the urgent need of real-time and continuous, yet non-invasive and automated monitoring of PVA, it is particularly critical to develop artificial intelligence technology that can automatically and accurately identify PVA. In contrast, Machine Learning (ML) methods offer advantages by learning from annotated data without pre-set thresholds, capturing complex relationships for high-accuracy classification. ML models are flexible and efficient, particularly with hardware acceleration, making them well-suited for real-time PVA monitoring [10].

This systematic review aims to collect and analyze studies employing ML techniques for PVA identification or prediction during MV. Identifying PVA allows immediate intervention, while prediction enables proactive adjustments to potentially prevent events and complications like VILI. ML technologies promise enhanced accuracy and efficiency for continuous, comprehensive PVA monitoring. By synthesizing the literature and assessing study quality, this review provides an overview of the field and aims to guide future research towards developing high-quality ML models for clinical decision-making support.

Machine learning model terminology

This section (Supplementary. Table 1) provides an overview of key terminology associated with the ML models utilized in the studies included in this review. Given that many medical professionals may not have extensive computer science backgrounds, the explanation will be presented in as straightforward and accessible a manner as possible to facilitate rapid understanding by the readers.

Materials and methods

Reporting

This study followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) reporting guidelines and is registered on the International Prospective Register of Systematic Reviews (PROSPERO identifier CRD42024626163).

Literature search

The following databases were searched systematically: (a) PubMed, (b) Embase, (c) Web of Science and (d) Cochrane, for articles published from inception to November 2024 using terms associating ML techniques with PVA. The complete filter is available in the Supplementary. Table 2. In addition, a manual search was also conducted on Google Scholar to find potential articles cited by the collected papers.

Eligibility criteria

In this study, The Population (P) was defined as adult patients undergoing MV. The Intervention (I) focused on the application of ML techniques. No Comparator (C) was included, as the review aimed to describe and evaluate the performance and application characteristics of ML models themselves. The Outcomes (O) were specific metrics related to the application and performance of ML for PVA. Study Designs (S) included original research studies.

Based on the defined PICOS framework and the study objectives, studies met the inclusion criteria if they involved adult patients undergoing MV, applied ML techniques, reported on at least one of the predefined outcomes related to the application and performance of ML

for PVA, and were published as original research studies in English. Conversely, studies were excluded if they did not involve adult patients on MV, did not apply ML specifically to PVA, were purely theoretical or methodological studies on ML without application to PVA data or patients, or were non-original research such as editorials, letters, conference abstracts, or review articles.

Data extraction and analysis

Two researchers (Jiang Guanxu and Xu Hao) independently screened titles and abstracts of the identified articles. Then, they obtained and read the full texts of potential articles; Zhu Zigang made the final decision in cases of disagreement. All processes during primary and secondary screening were supervised by Ma Jiawei.

Data extracted from the included articles encompassed study characteristics, such as year of publication, sample size, and breath volume. Information specific to the machine learning models was also collected, including the type of data used, the specific ML methods employed, and reported performance metrics (e.g., accuracy, sensitivity, specificity, F1 Score, and area under the curve [AUC]). Furthermore, details relevant to the quality assessment were systematically retrieved. These included information on the use of hyperparameters, details of internal and external validation, outcome assessment, strategies for addressing class imbalance, feature selection, data processing methods, and handling of missing data.

Quality assessment

For prediction studies, we assessed the risk of bias using the Prediction Model Risk of Bias Assessment (PROBAST) [11] tool. This tool systematically examines four essential domains: participants, predictors, outcome, and analysis, to identify potential sources of bias in prediction model studies.

However, while a standardized tool like PROBAST exists for assessing prediction models, the academic community currently lacks a single universally accepted and comprehensive quality assessment tool specifically designed for systematic reviews of ML classification studies, particularly within clinical domains involving complex physiological time-series data like PVA. To address this gap and ensure a robust evaluation of the included classification studies, we developed a refined quality assessment method. Our method builds upon the framework proposed by Ramos-Lima et al. (LansA team) [12], which offers a valuable foundational approach for assessing ML studies in trauma-related disorders.

Upon careful consideration of the original LansA framework, we identified a need for more granular and specific criteria relevant to the intricacies of ML model development and evaluation in physiological signal analysis, such as those encountered in PVA. Specifically, based on a thorough review of best practices in machine learning methodology and the specific challenges inherent in PVA datasets (e.g., highly imbalanced classes, complex time-series data, and the critical need for robust clinical generalizability), we refined and expanded several key domains. The specific modifications from the original LansA framework and their rationale are in Supplementary. Table 3.

Our improved version systematically evaluates nine key domains: representativeness of the sample, data Processing, feature selection, machine learning approach, Performance/Accuracy, Internal/External validation, class imbalance, Overfitting and Underfitting, and Hyperparameter Tuning. Table 1 describes the domains in details. To further ensure the methodological rigor and appropriateness of our refined quality assessment method, it was reviewed and approved by Dr. YanTao, a co-author and expert in Artificial Intelligence from Jiangnan University. His expertise provided an additional layer of validation for the chosen domains, ensuring their relevance and comprehensiveness for evaluating ML classification studies.

Table 1 Quality assessment for machine learning classification studies

Feature	Considerations
1. Representativeness of the sample	Was the study truly representative of the target population heterogeneity? If not, was this related to the selected sampling method, insufficient sample size or inclusion/exclusion criteria?
2. Data Processing	Is there a clear and explicit process for data filtering and annotation? Are the category labels accurate and consistent?
3. Feature Selection	Is feature selection based on reasonable scientific evidence or prior knowledge? Are feature definitions clear and consistent? Are feature measurement methods accurate and reliable? If the feature is automatically extracted by machine algorithm, it is marked as "-".
4. Machine Learning Approach	Was the machine learning algorithm used to analyze the data clearly described and appropriate?
5. Performance/Accuracy	Were multiple performance metrics used to comprehensively quantify model performance? (e.g., Accuracy, Sensitivity, Specificity, AUC, PPV/NPV)
6. Internal/External validation	Was internal validation performed to assess model stability? Was external validation conducted to evaluate model generalizability?
7. Class Imbalance	Did the authors address the issue of class imbalance?
8. Overfitting or Underfitting	Were overfitting or underfitting appropriately addressed and considered?
9. Hyperparameter Tuning	Was hyperparameter tuning described?

The quality assessment for all included studies was independently performed by two researchers (Jiang-Guanxu and LuoLiang). Any disagreements were resolved through discussion, and if consensus could not be reached, a third reviewer (YanTao) made the final decision.

Results

Study selection

We found a total of 74 potential abstracts and included 14 articles in the present review. Figure 1 presents the literature search results, displaying the number of studies screened, assessed for eligibility, and reviewed, along with the reasons for exclusions at each stage. A list of the included articles as well as the most relevant characteristics and findings are presented in Table 2 (Classification studies), Table 3 (Prediction studies), and Supplementary. Table 3 (Technical details of machine learning model in Included Studies).

Study characteristics

Fourteen included studies enrolled 469 adult patients with MV. Thirteen of these studies utilized current information to identify different types of PVA, while only one [25] study predicted the occurrence of PVA at a future time point. Two [17, 24] of the studies were prospective, the remaining 12 were retrospective. The research period primarily spanned from 2018 to 2024, involving a total of 11 ML methods and 19 targeted types of PVA in mechanical ventilation, with a breath volume reaching 10 million breaths. One [14] of the studies employed a simulation method to generate respiratory waveforms, aiming to mitigate the adverse effects of small sample sizes on machine learning modeling.

Machine learning models characteristics and performance

As depicted in Supplementary Table 1, eleven different ML models were employed to identify or predict different types of PVA. These models ranged from traditional algorithms such as RF and SVM to more advanced deep learning architectures including CNN and LSTM. CNN and its variants (CNN-LSTM, Modified U-net CNN, 1DCNN, TCNN) were the most common ML models mentioned in 7 studies, followed by RF in 4 studies. In addition to using ML methods independently, researchers have also explored the combination of multiple ML methods in an attempt to enhance model performance. Chen et al. [13] employed an Attention-based CNN-LSTM, which leverages the strengths of both CNN and LSTM, effectively capturing temporal features from airway pressure and flow waveform data. Rehm et al. [15] utilized ensemble learning methods, including RF, MLP, ERTC, and GBC, to identify Double Triggering and Breath Stacking, further improving the performance of the model.

Regarding the evaluation of ML model performance, different studies evaluate the effect of ML model through a variety of evaluation indicators. Common evaluation indexes include accuracy, sensitivity, specificity, F1 score, MCC, AUROC, and RMSE. As the most commonly reported evaluation metric, F1 score provides a balanced measure of precision and recall, making it particularly suitable for imbalanced datasets. Across the 8 studies that reported F1 scores, the values ranged from 0.731 [21] for delayed inspiration detection to 0.988 [13] for double triggering detection. From the results, it seems like that there was no clear correlation between model complexity and performance. While deep learning models such as CNN and LSTM were commonly used and often achieved high performance, traditional algorithms like RF and SVM also demonstrated promising results in some studies.

It is worth noting that although different researches have differences in ML models and feature selection, most of them emphasize the importance of model calibration, hyper parameter optimization and verification process. Most studies used cross-validation to evaluate model performance, but only a few conducted external validation using an independent dataset. This highlights the need for future studies to validate their models on external datasets to ensure generalizability.

Quality assessment

The results of the quality assessment are presented in Tables 4 and 5. In classification studies (Table 4), regarding the representativeness of the samples, approximate half of the studies (7 of 13 studies; 53.8%) considered a statistical sample or a larger sample that was representative of the described population of the study. The majority of the studies (10 of 13 studies; 76.9%) provided a detailed description of the processes of data collection, annotation, and feature extraction. The feature extraction methods were scientific and reasonable, and the feature labels were annotated by clinical professionals, ensuring high accuracy and consistency. Due to the different ML methods employed in the studies, 9 of 13 articles involved automatic feature extraction using algorithms, while the majority of the remaining studies (3 of 4 studies) discussed in detail about the process of feature selection by using expert knowledge and statistical methods. Almost all of the studies (12 of 13 studies; 92.3%) clearly described the ML algorithms used, and some even provided sufficient details about the algorithms and experimental settings to make the model-building process reproducible. The majority of the studies (10 of 13 studies; 76.9%) employed multiple performance metrics, such as accuracy, sensitivity, specificity, and F1 score, to

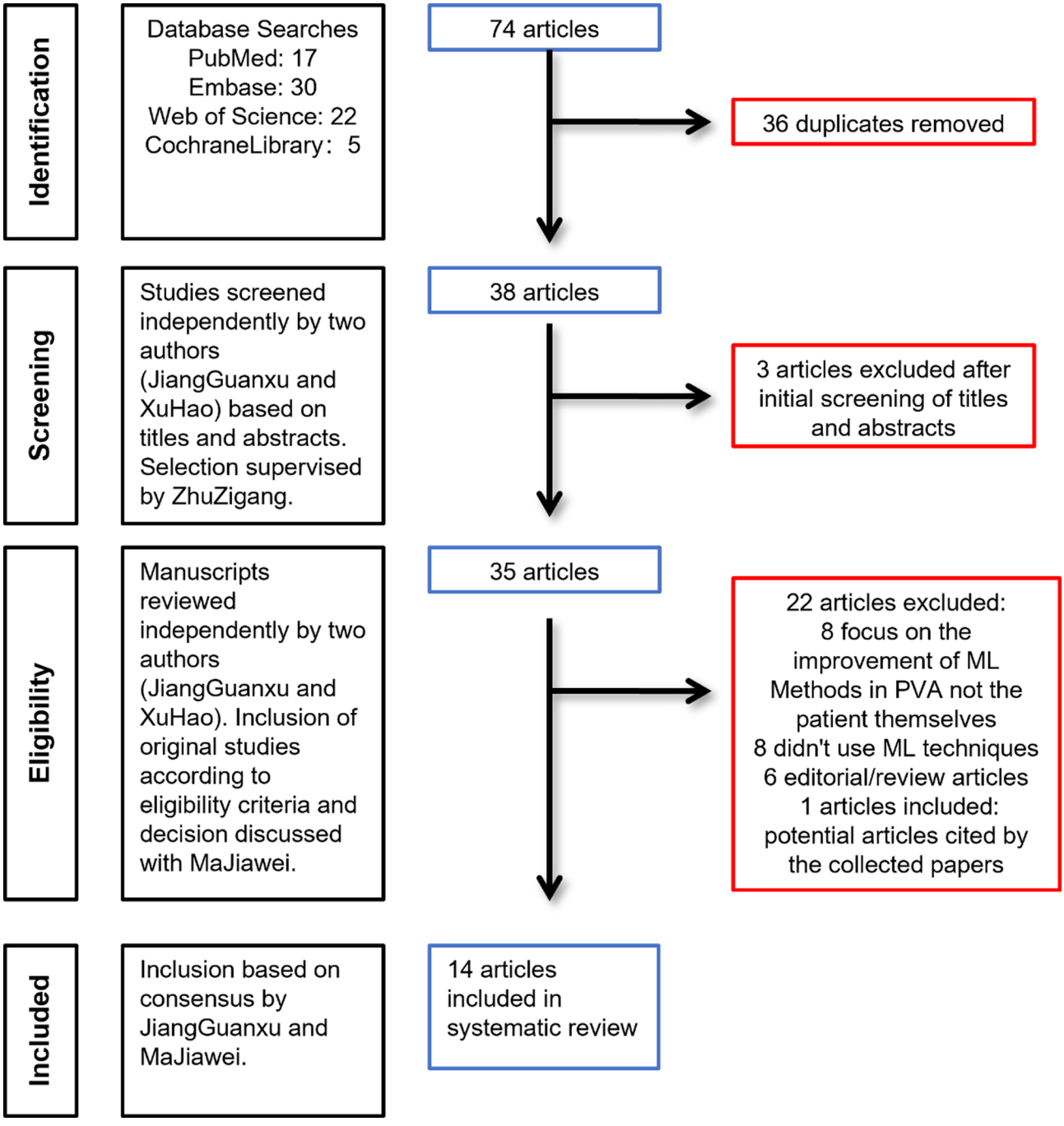

Fig. 1 Flow chart of the proposed literature search. ML: Machine learning. PVA: Patient-ventilator asynchrony

quantify model performance and presented detailed classification results. Almost all of the studies (12 of 13 studies; 92.3%) has undergone internal validation to assess the stability of the models, but only two studies have conducted external validation to evaluate the models' generalization ability. Regarding the technical details in the modeling process, 5 studies have addressed the issue of class imbalance, 8 articles have considered the problems of overfitting or underfitting and reduced the associated risks, and 4 studies have provided a detailed description of the hyperparameter tuning process to ensure optimal model performance. In Prediction studies (Table 5), only one study analyzed using the PROBAST tool for risk of bias assessment, this study was found to have a low risk of bias.

Table 2 Classification studies

Author	Sample Size	MV mode	Breath volume (breaths)	ML methods employed	Target asynchrony	Characteristics of ML model modeling	Performance
Chen et al. [13]	10	PCV	≈30,000	Attention-based CNN-LSTM	IE, DT	Airway pressure and flow waveform data	Mean Accuracy: 0.989 Sensitivity: IE: 0.991, DT: 0.993 Specificity: IE: 0.967, DT: 0.972 F1 Score: IE: 0.986, DT: 0.988 MCC: IE: 0.918, DT: 0.927
Bakkes et al. [14]	15	PSV	4275 + 58,876(s)	Modified U-net CNN	DI, LC, EC, IE	Timing information of patients' breaths	Overall accuracy > 90% F1 Score: DI-79.9%, LC- 92.5% EC-88.3%, IE-91.4%
Rehm et al. [15]	35	-	9719	Ensemble learning (including RF, MLP, ERTC, GBC)	DT, BS	TVi (Inspiratory Tidal Volume), TVe (Expiratory Tidal Volume), TVe/TVi (Ratio of Expiratory to Inspiratory Tidal Volume), I-time (Inspiratory Time), E-time (Expiratory Time), I:E ratio (Inspiratory to Expiratory Time Ratio), RR (Instantaneous Respiratory Rate), etc.	Sensitivity: DT: 0.960,BS:0.944 Specificity: DT: 0.975,BS:0.987
Zhang et al. [16]	159	PC AC	185,385/ 330,825	LSTM Neural Network	IE, DT	Temporal features extracted from pressure and flow waveforms	Sensitivity: DT: 0.988, IEE: 0.988 Specificity: DT: 0.979, IEE: 0.971 F1 Score: DT: 0.983, IEE: 0.979
DeHaro et al. [17]	28	A/C	6428	RNN, CNN	Flow starvation	Airway pressure waveform (Paw)	Accuracy: RNN:87.9%;CNN: 86.8% F1 Score: RNN:87.7%;CNN: 86.9%
Pan et al. [18]	20	A/C	516,210	CNN (based on pretrained MobileNet, VGG 16, Inception-ResNetV2 CNNs)	IE, DT	2D image features converted from airway pressure, flow rate, and tidal volume sequences	F1 Score: (Pr_CNN_SVM model, PRR = 0.1) DT: Up to 0.965 IE: Up to 0.949
Pan et al. [19]	20	PCV PSV	108,263 breaths (PCV mode) 180,996 breaths (PSV mode)	1DCNN	IE, DT, PC, DC	Airway pressure and flow waveform data	Sensitivity: IE>0.985, DT>0.989 PC>0.988, DC>0.978 Specificity: IE>0.959, DT>0.967 PC>0.954, DC>0.966 F1 Score: IE>0.971, DT>0.980 PC>0.969, DC>0.973
Ge et al. [20]	7	-	3,923,450	a two-layer LSTM	IE, DT	Airway pressure and flow waveforms	Accuracy > 95%

Table 2 (continued)

Author	Sample Size	MV mode	Breath volume (breaths)	ML methods employed	Target asynchrony	Characteristics of ML model modeling	Performance
Bakkes et al. [21]	15	-	4275	1DCNN	DI, LC, EC, IE	Air pressure, flow, and volume curve signals	Inspiratory effort detection: Sensitivity 98.6%, Precision 97.3% Expiratory effort detection: Sensitivity 97.7%, Precision 97.2% F1 Score: DI:73.1%,LC:92.6% EC:87.1%,IE:95.0%
Zhang et al. [22]	25	-	≈20,000	RF, SVM, KNN	Missed triggering, DT, prolonged cycling, circuitry condensat-ion, expiratory flow limit	Time-domain features (e.g., maximum value, minimum value, standard deviation), Time-frequency features (e.g., mean of absolute value, average power from wavelet decomposition), Entropic features (e.g., fuzzy entropy, approximate entropy)	F1 Score: Prolonged cycling 97.56%, Double triggering 92.26%, Missed triggering 96.46%, Circuitry condensation 89.18%, Expiratory flow limit 96.05%
Ang et al. [9]	11	VC PC	56,000	TCNN	FA, RT, PC, DT, DC, IE, AT	Flow, pressure, and volume waveform data	Accuracy: 89.5%, Sensitivity: 98.3%, Specificity: 83.9%
Gholami et al. [23]	11	PC	1,204	RF	DT*, PT	Pressure "tenting", flow reversal, estimated inspiratory time, ratio of negative flow time to inspiratory time	Sensitivity: DT: 94%, PT: 89% Specificity: DT: 98%, PT: 99%
Sottile et al. [24]	62	APVCMV PC	4,256,323	RF, AdaBoost	DT, FL, IT	Features based on ventilator pressure, flow, and volume measurements, along with temporal characteristics of these data.	AUC-ROC: DT: > 0.89 FL: = 0.89 IT: = 0.80

Breath volume: (s): Simulation generated data. Machine ventilation mode: (1) PCV: Pressure Control Ventilation; (2) PSV: Pressure Support Ventilation; (3) VC: Volume Control; (4) PC: Pressure Control; (5) A/C: Assist/Control Ventilation; (6) APVCMV: Assisted Pressure Control with Volume Cycled Mandatory; (7) AC: Assist Control. Machine learning methods employed: (1) CNN-LSTM: Convolutional Long Short-Term Memory Neural Network; (2) CNN: Convolutional Neural Network; (3) RF: Random Forest; (4) MLP: Multilayer Perceptron; (5) ERTC: Extremely Randomized Trees classifier; (6) GBC: Gradient Boosted classifier; (7) LSTM: Two-layer Long Short-Term Memory; (8) RNN: Recurrent Neural Network; (9) SVM: Support vector machine; (10) 1DCNN: One-Dimensional Convolutional Neural Network; (11) KNN: k-Nearest Neighbors; (12) TCNN: Tri-input Convolutional Neural Network. Target asynchrony: (1) IE: Ineffective effort; (2) DT: Double Triggering; (3) DI: Delayed Inspiration; (4) LC: Late Cycling; (5) EC: Early Cycling; (6) BS: Breath Stacking; (7) FA: Flow Asynchrony; (8) RT: Reverse Triggering; (8) PC: Premature Cycling; (9) DC: Delayed Cycling; (10) AT: Auto Triggering; (11) FL: Flow Limitation; (12) IT: Ineffective Triggering; (13) DT*: Delayed Termination; (14) PT: Premature Termination. Performance: (1) MCC: Matthews Correlation Coefficient; (2) TPR: True positive rate; (3) TNR: True negative rate; (4) PRR: Partial retention rate; (5) AUC-ROC: Area Under the Receiver Operating Characteristic Curve

Table 3 Prediction studies

Author(Year)	Sample size	Breath volume(breaths)	ML methods employed	Target asynchrony	Characteristics of ML model modeling	Performance
Marchuk et al. [25]	51	10,409,357	Hidden Markov Model	IE, DC, SC, PC	Time series data representing the total count of asynchrony events	RMSE: 19.5 (Base model:35.4)

ML: Machine learning; Target asynchrony: (1) IE: Ineffective effort; (2) DC: Double cycling; (3) SC: Short cycling; (4) PC: Prolonged cycling; Performance; (1) RMSE: The root mean square error

Discussion

PVA remains a significant challenge in MV, often requiring labor-intensive manual assessment by skilled clinicians. ML offers a promising avenue for automated and accurate PVA detection and prediction, potentially overcoming the limitations of current methods. To our knowledge, this is the first systematic review to comprehensively evaluate the methodological and technical aspects of ML applications in PVA, providing a critical overview of the current landscape and identifying key areas for future development. We identified and analyzed 14 original studies, predominantly published in recent years, focusing on the methodologies and performance of ML models for identifying various PVA types

Table 4 Quality assessment tool for classification studies

Author	Sample Representativeness	Data Processing	Feature Selection	ML Approach	Performance/Accuracy	Internal/External validation	Class Imbalance	Overfitting or Underfitting	Hyperparameter Tuning
Chen et al. [13]	√	√	-	√	√	√/×	×	√	√
Bakkes et al. [14]	×	√	-	√	√	√/×	×	√	×
Rehm et al. [15]	√	×	√	√	√	√/×	√	√	×
Zhang et al. [16]	√	×	-	√	√	√	√	√	√
DeHaro et al. [17]	×	√	-	√	√	√/×	×	×	×
Pan et al. [18]	√	√	-	√	×	√/×	×	√	×
Pan et al. [19]	×	√	-	√	√	√/×	√	√	×
Ge et al. [20]	×	√	-	×	×	×	×	×	×
Bakkes et al. [21]	×	√	-	√	√	√/×	×	√	×
Zhang et al. [22]	√	√	√	√	√	√/×	√	×	×
Ang et al. [9]	√	×	-	√	√	√	√	√	×
Gholami et al. [23]	×	√	√	√	√	√/×	×	×	×
Sottile et al. [24]	√	√	×	√	×	√/×	×	×	×

Symbol annotations: √: Meets our quality evaluation criteria. ×: Does not meet quality evaluation criteria, or not mentioned, or unclear. -: Features automatically extracted by machine learning. ML: Machine learning

Table 5 Risk of Bias assessment results using PROBAST tool for prediction studies of included studies

Author	Domain 1 (Participants)		Domain 2 (Predictors)		Domain 3 (Outcomes)		Domain 4 (Analysis)	Overall judgement	
	Risk of Bias	Applicability	Risk of Bias	Applicability	Risk of Bias	Applicability		Risk of Bias	Concern for Applicability
Marchuk et al. [25]	Low	Low	Low	Low	Low	Low	Low	Low	Low

or predicting their occurrence. While these studies, generally assessed as having a low risk of bias for the single prediction study or meeting most quality criteria for classification studies via our refined method, and collectively highlight the significant potential of ML, a formal meta-analysis is still not feasible. This decision was primarily due to the substantial heterogeneity across the included studies concerning the specific ML models utilized, the diverse types of PVA investigated, the variability in data sources and characteristics, and the lack of standardized evaluation metrics, which precluded meaningful quantitative synthesis.

Our review of classification studies revealed a notable heterogeneity in the reported evaluation metrics, with common measures including accuracy, sensitivity, specificity, and F1 score (Table 2). While these metrics offer valuable insights, their utility is limited in the context of inherent class imbalance prevalent in PVA datasets, where asynchrony instances are significantly less frequent than normal breaths. In such scenarios, overall accuracy can be misleading, and even F1 score, while balancing precision and recall, may not fully capture a model's effectiveness for the minority (asynchrony) class. More robust metrics, such as the Matthews Correlation Coefficient (MCC) and Area Under the Receiver Operating Characteristic Curve (AUROC), are crucial as they provide a more reliable assessment by considering all four quadrants of the confusion matrix. However, our review found that AUROC and MCC were inconsistently reported across studies (only two studies [13, 24] included them), significantly hindering comprehensive performance comparison. Furthermore, no clear correlation between model complexity and performance was consistently evident, likely due to variations in datasets, targeted PVA types, and evaluation methodologies. Therefore, we strongly recommend that future classification studies prioritize the consistent use of robust metrics like AUROC and MCC, alongside traditional measures, to ensure a more complete and objective assessment of model performance, especially in the presence of class imbalance.

Additionally, there is a lack of standardization in how PVA is defined and measured across different studies, which hinders the comparison and validation of ML models [26]. Although some studies have attempted to categorize the different types of PVA [27], it is evident that not all types of PVA reported in the historical literature can be included. Some studies may refer to PVA with slightly different names but convey similar meanings, such as the Ineffective Effort (IE) and Ineffective Triggering (IT) summarized in this study, leading to repeated research on similar types of PVA by different research teams and resulting in a waste of resources. Currently, a unified and comprehensive classification system for PVA is urgently needed. We advocate for an international expert consensus meeting to review all known PVA types, establish clear and standardized definitions, and recommend comprehensive outcome measures for their assessment. Such a consensus would provide researchers with well-defined targets, streamline model development, facilitate robust comparisons, and ultimately accelerate the translation of accurate and reliable ML tools into clinical practice.

While significant progress has been made in identifying existing PVA events, which is crucial for immediate intervention, the ability to predict the occurrence of PVA in advance holds distinct clinical appeal. Proactive adjustments to ventilator settings based on predicted risk could potentially prevent asynchrony, improve patient comfort, and reduce associated complications like VILI. However, our systematic search revealed a notable scarcity of research in this area, identifying only one study [25] specifically focused on predicting the occurrence of PVA at a future time point. Marchuk et al. [25] demonstrated the feasibility of risk prediction using a Hidden Markov Model to categorize patients into varying risk levels based on asynchrony event counts. While this study showed promise, the clinical utility of intervening based on these predicted risk levels, particularly in terms of improving patient outcomes, remains an open question. This significant imbalance between classification and prediction research highlights a critical gap in the current literature and underscores the need for more investigations into ML models capable of forecasting PVA and assessing the clinical impact of such predictive approaches.

The choice of MV mode is another critical factor that warrants careful consideration in the development and application of ML models for PVA. Different ventilation modes impose distinct interactions between the patient and the ventilator, which in turn significantly influences the types and frequency of PVA events observed, and consequently, the characteristics of the physiological signals (pressure, flow, and volume) used for ML model training. For example, ineffective triggering and double

triggering are more common in Pressure Support Ventilation (PSV) mode, while insufficient flow and reverse triggering are more prominent in Volume Controlled Ventilation (VCV) mode [28]. Models trained exclusively on data from one mode may not perform optimally or generalize well when applied to data from another. While most included studies reported the ventilation modes from which their data were sourced, few explicitly addressed how their models' performance might vary across modes or demonstrated generalizability. Therefore, future ML models aimed at widespread clinical application must demonstrate robustness and adaptability across diverse ventilation modes. This may necessitate training models on datasets encompassing multiple ventilation modes or developing adaptive algorithms capable of accounting for mode-specific signal characteristics to ensure reliable performance regardless of the ventilation strategy employed.

Among the 14 included studies, despite the limitations in outcome measures as mentioned above, the performance of the models is encouraging. From the results, it appears that combining different ML methods for modeling achieves the best performance in the field of PVA research. For instance, Chen et al. [13] combined the CNN-LSTM method to identify IE, DT, and MCC all above 90%, but this study was only validated internally, leaving the model's generalization ability uncertain. In the two studies [9, 16] that conducted external validation, the ML methods used were still CNN and LSTM, which achieved good results, indirectly indicating the greater potential of CNN and LSTM in the field of PVA identification. Future research should continue to explore the potential of hybrid or ensemble ML approaches while prioritizing rigorous external validation on independent datasets to confirm their real-world applicability.

Regarding technical details, our review revealed that while most studies considered the risk of overfitting, detailed reporting on methods used to address class imbalance and the settings of model hyperparameters was notably limited. Class imbalance, a prominent issue in PVA studies where asynchrony events are often minority classes, can bias models towards the majority class, impairing their ability to recognize critical minority events [29]. Only two studies [15, 16] explicitly utilized the SMOTE algorithm to address this. Similarly, hyperparameter tuning, crucial for optimizing model performance and ensuring generalizability, was often inadequately described. This lack of transparency regarding class imbalance handling and hyperparameter settings significantly impedes the reproducibility of reported results and makes it challenging for other researchers to replicate models or compare their work against reported benchmarks.

While ML model performance is influenced by various factors, sample size emerges as a particularly crucial determinant [30], especially given the challenges in obtaining large, high-quality labeled datasets in clinical settings. Our review found that despite these constraints, some included studies have employed innovative approaches to mitigate the limitations imposed by small sample sizes, offering valuable insights for future research teams.These strategies primarily fall into two categories:

Leveraging advanced learning paradigms: Researchers [18] have utilized methods like image transfer learning to achieve high accuracy in PVA recognition even with limited data. This approach transforms one-dimensional physiological signals into two-dimensional images, allowing pre-trained deep learning models to extract features effectively, demonstrating robust performance even when data volume is significantly reduced.

Generating synthetic data: To compensate for data scarcity, studies have explored generating data similar to real labeled data. This includes model-based simulations [14, 31], where nonlinear lung and airway models are used to generate large volumes of labeled respiratory waveform data that closely mimic clinical observations. Additionally, adversarial learning methods [32], such as Generative Adversarial Networks (GANs), have been employed to synthesize realistic patient-ventilator waveforms while preserving specific labels, significantly improving classification accuracy for PVA detection models trained on such augmented datasets.

These innovative approaches—including transfer learning, model-based simulation, and adversarial learning—provide valuable references for future research facing small sample sizes, demonstrating pathways to enhance model quality and generalizability despite data limitations.

Conclusion

This systematic review demonstrates the significant potential of ML models for the automated identification and prediction of PVA. Despite promising performance reported in individual studies, their widespread clinical application is currently limited by critical challenges. These include substantial methodological heterogeneity, a lack of standardized definitions and comprehensive outcome measures for PVA types, and crucially, insufficient external validation of developed models in diverse clinical settings. To realize the full clinical benefit of ML in PVA management, future efforts must focus on establishing uniform standards, enhancing methodological rigor and reporting (particularly regarding technical details like class imbalance and hyperparameter tuning), and conducting large-scale, multicenter prospective studies with robust external validation.

Abbreviations

PVA	Patient-ventilator asynchrony
ML	Machine learning
CNNs	Convolutional Neural Networks
MCC	Matthews Correlation Coefficient
RMSE	The root mean square error
MV	Mechanical ventilation
ICU	Intensive Care Unit
VILI	Ventilator-associated lung injury
RBMs	Rule-Based Methods
PICOS	Population, Intervention, Comparator, Outcomes, Study Designs
AUC	Area Under the Curve
HMM	Hidden Markov Model
PSV	Pressure Support Ventilation
VCV	Volume Controlled Ventilation
IE	Ineffective Effort
DT	Double Triggering
IT	Ineffective Triggering
GAN	Generative Adversarial Network

Supplementary Information

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Supplementary Material 1: Supplementary Table 1. A summary of the ML models presented in the papers in our survey. Supplementary Table 2. Employed databases and search strategies. Supplementary Table 3. The specific modifications from the original LansA framework and their rationale. Supplementary Table 4. Technical details of machine learning model in Included Studies

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Author contributions

Guanxu Jiang is the first author and contributed to the study design, data extraction, analysis, and manuscript writing. Liang Luo is the first corresponding author and oversaw the overall study, with responsibility for reviewing and validating the medical aspects of the research. Tao Yan is the second corresponding author and provided expertise in reviewing and validating the computer science and machine learning aspects of the research. Jiawei Ma supervised all processes during primary and secondary screening. Hao Xu independently screened titles and abstracts of the identified articles and extracted data. Zigang Zhu made the final decision in cases of disagreement during article screening. Yi Xie and Shenhui Ji contributed to the data extraction and analysis. All the authors read and approved the final manuscript.

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Data availability

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Declarations

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Competing interests

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